

Name:- Sejal Rollno :- S006 Class:- SYIT Subject :-PDS

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) StudentPerf..Factors.csv

StudentPerformanceFactors.csv(text/csv) - 641949 bytes, last modified: 9/30/2026 - 100% done
Saving StudentPerformanceFactors.csv to StudentPerformanceFactors.csv

```
import pandas as pd
```

Q1 Import a CSV dataset using Pandas.

```
data = pd.read_csv("StudentPerformanceFactors.csv")
```

```
print(data.head())
```

```

Hours_Studied  Attendance  Parental_Involvement  Access_to_Resources  \
0             23           84                  Low                High
1             19           64                  Low                Medium
2             24           98                Medium                Medium
3             29           89                  Low                Medium
4             19           92                Medium                Medium

Extracurricular_Activities  Sleep_Hours  Previous_Scores  Motivation_Level  \
0                        No              7              73                Low
1                        No              8              59                Low
2                        Yes              7              91                Medium
3                        Yes              8              98                Medium
4                        Yes              6              65                Medium

Internet_Access  Tutoring_Sessions  Family_Income  Teacher_Quality  \
0             Yes                  0              Low                Medium
1             Yes                  2              Medium                Medium
2             Yes                  2              Medium                Medium
3             Yes                  1              Medium                Medium
4             Yes                  3              Medium                High

School_Type  Peer_Influence  Physical_Activity  Learning_Disabilities  \
0     Public      Positive              3                No
1     Public     Negative              4                No
2     Public     Neutral              4                No
3     Public     Negative              4                No
4     Public     Neutral              4                No

Parental_Education_Level  Distance_from_Home  Gender  Exam_Score
0             High School              Near  Male           67
1             College              Moderate  Female          61
2             Postgraduate              Near  Male           74
3             High School              Moderate  Male           71
4             College              Near  Female           70

```

Q2.Find missing values in the dataset.

```
print(data.isnull().sum())
```

```

Hours_Studied           0
Attendance              0
Parental_Involvement    0
Access_to_Resources     0
Extracurricular_Activities  0
Sleep_Hours             0
Previous_Scores         0
Motivation_Level        0
Internet_Access         0
Tutoring_Sessions       0
Family_Income           0
Teacher_Quality        78
School_Type            0
Peer_Influence          0
Physical_Activity       0
Learning_Disabilities   0
Parental_Education_Level  90
Distance_from_Home      67

```

```
Gender          0
Exam_Score      0
dtype: int64
```

Q3. Replace missing numerical values with the mean.

```
data.fillna(data.mean(numeric_only=True), inplace=True)
print(data)
```

```

      Hours_Studied  Attendance  Parental_Involvement  Access_to_Resources  \
0                23           84                Low           High
1                19           64                Low           Medium
2                24           98                Medium           Medium
3                29           89                Low           Medium
4                19           92                Medium           Medium
...            ...           ...                ...           ...
6602             25           69                High           Medium
6603             23           76                High           Medium
6604             20           90                Medium           Low
6605             10           86                High           High
6606             15           67                Medium           Low

      Extracurricular_Activities  Sleep_Hours  Previous_Scores  \
0                          No              7              73
1                          No              8              59
2                         Yes              7              91
3                         Yes              8              98
4                         Yes              6              65
...            ...           ...           ...
6602                      No              7              76
6603                      No              8              81
6604                      Yes              6              65
6605                      Yes              6              91
6606                      Yes              9              94

      Motivation_Level  Internet_Access  Tutoring_Sessions  Family_Income  \
0                Low              Yes              0              Low
1                Low              Yes              2              Medium
2                Medium           Yes              2              Medium
3                Medium           Yes              1              Medium
4                Medium           Yes              3              Medium
...            ...           ...           ...           ...
6602             Medium           Yes              1              High
6603             Medium           Yes              3              Low
6604                Low           Yes              3              Low
6605                High           Yes              2              Low
6606             Medium           Yes              0              Medium

      Teacher_Quality  School_Type  Peer_Influence  Physical_Activity  \
0             Medium      Public      Positive              3
1             Medium      Public      Negative              4
2             Medium      Public      Neutral              4
3             Medium      Public      Negative              4
4              High      Public      Neutral              4
...            ...           ...           ...           ...
6602             Medium      Public      Positive              2
6603              High      Public      Positive              2
6604             Medium      Public      Negative              2
6605             Medium     Private      Positive              3
6606             Medium      Public      Positive              4

      Learning_Disabilities  Parental_Education_Level  Distance_from_Home  \
0                      No              High School              Near
1                      No              College              Moderate
2                      No              Postgraduate              Near
3                      No              High School              Moderate
4                      No              College              Near
```

Q4. Remove duplicate records.

```
data.drop_duplicates(inplace=True)
print(data)
```

2	Medium	Yes	2	Medium
3	Medium	Yes	1	Medium
4	Medium	Yes	3	Medium
...	...	...	...	...
6602	Medium	Yes	1	High
6603	Medium	Yes	3	Low
6604	Low	Yes	3	Low
6605	High	Yes	2	Low
6606	Medium	Yes	0	Medium

	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	\
0	Medium	Public	Positive	3	
1	Medium	Public	Negative	4	
2	Medium	Public	Neutral	4	
3	Medium	Public	Negative	4	
4	High	Public	Neutral	4	
...	...	...	...	...	
6602	Medium	Public	Positive	2	
6603	High	Public	Positive	2	
6604	Medium	Public	Negative	2	
6605	Medium	Private	Positive	3	
6606	Medium	Public	Positive	4	

	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	\
0	No	High School	Near	
1	No	College	Moderate	
2	No	Postgraduate	Near	
3	No	High School	Moderate	
4	No	College	Near	
...	...	...	...	
6602	No	High School	Near	
6603	No	High School	Near	
6604	No	Postgraduate	Near	
6605	No	High School	Far	
6606	No	Postgraduate	Near	

	Gender	Exam_Score
0	Male	67
1	Female	61
2	Male	74
3	Male	71
4	Female	70
...	...	...
6602	Female	68
6603	Female	69
6604	Female	68
6605	Female	68
6606	Male	64

[6607 rows x 20 columns]

Q5.Display the cleaned dataset.

```
print("Cleaned Dataset:")
print(data)
```

Cleaned Dataset:

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	
...	...	...	...	...	
6602	25	69	High	Medium	
6603	23	76	High	Medium	
6604	20	90	Medium	Low	
6605	10	86	High	High	
6606	15	67	Medium	Low	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	\
0	No	7	73	
1	No	8	59	
2	Yes	7	91	
3	Yes	8	98	
4	Yes	6	65	
...	...	...	...	
6602	No	7	76	
6603	No	8	81	
6604	Yes	6	65	
6605	Yes	6	91	
6606	Yes	9	94	

	Motivation_Level	Internet_Access	Tutoring_Sessions	Family_Income	\
0	Low	Yes	0	Low	
1	Low	Yes	2	Medium	
2	Medium	Yes	2	Medium	
3	Medium	Yes	1	Medium	
4	Medium	Yes	3	Medium	
...	...	...	...	...	
6602	Medium	Yes	1	High	
6603	Medium	Yes	3	Low	
6604	Low	Yes	3	Low	
6605	High	Yes	2	Low	
6606	Medium	Yes	0	Medium	

	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	\
0	Medium	Public	Positive	3	
1	Medium	Public	Negative	4	
2	Medium	Public	Neutral	4	
3	Medium	Public	Negative	4	
4	High	Public	Neutral	4	
...	...	...	...	...	
6602	Medium	Public	Positive	2	
6603	High	Public	Positive	2	
6604	Medium	Public	Negative	2	
6605	Medium	Private	Positive	3	
6606	Medium	Public	Positive	4	

	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	\
0	No	High School	Near	
1	No	College	Moderate	
2	No	Postgraduate	Near	
3	No	High School	Moderate	